DAY 20 – LINUX USER MANAGEMENT

Cyber Security Internship – Veda Technology

1. Objective

To understand the fundamentals of Linux user and group management, create user accounts, assign users to groups, and manage directory ownership and access permissions in a safe Linux practice environment.

2. Introduction

Linux is a multi-user operating system that allows multiple users to access and work on the same system. Proper user management is essential for maintaining security, protecting sensitive information, and preventing unauthorized access.

Linux provides several commands for creating, modifying, and managing user accounts and groups. Each user has a unique User ID (UID), and each group has a Group ID (GID).

User and group management helps administrators control access to system resources by assigning appropriate permissions.

Key Concepts

Concept	Description
User	An individual account used to access the Linux system
Group	A collection of users sharing access privileges
UID	Unique identification number assigned to a user
GID	Unique identification number assigned to a group
Root User	Administrator account with extensive system privileges
File Ownership	Identifies the user and group associated with a file
Permissions	Define read, write, and execute access

3. Linux User Management Commands

The following commands are commonly used for managing Linux users, groups, and access permissions.

Command	Purpose
whoami	Displays the currently logged-in username
getent passwd	Lists user account information
useradd	Creates a new user account
passwd	Sets or changes a user's password

groupadd	Creates a new group
usermod -aG	Adds a user to a supplementary group
id	Displays user and group identification details
mkdir	Creates a new directory
chown	Changes file or directory ownership
chmod	Modifies access permissions
ls -ld	Displays directory permissions and ownership

4. Practical Implementation

The following procedure demonstrates basic Linux user and group management using Ubuntu Terminal.

Step 1: Identify the Current User

Command:

whoami

Description:

The whoami command displays the username of the currently logged-in account. It helps identify which user is executing commands.

Illustrative output:

yuktha

Step 2: View Existing Linux Users

Command:

getent passwd

Description:

This command retrieves information about existing user accounts from the system's configured account databases.

The output includes usernames, user IDs, group IDs, home directories, and login shells.

Step 3: Create a New User

Commands:

sudo useradd -m intern1

sudo passwd intern1

Description:

The useradd command creates a new Linux account named intern1.

The -m option creates a home directory for the user, while passwd sets a password for the newly created account.

Expected result: A new user account named intern1 is created with a home directory.

Step 4: Create a New Group

Command:

sudo groupadd cybersecurity

Description:

The groupadd command creates a group named cybersecurity.

Groups allow administrators to assign common permissions to multiple users instead of managing each account individually.

Step 5: Add User to Group

Command:

sudo usermod -aG cybersecurity intern1

Description:

The usermod command modifies an existing user account.

- -a appends the user to supplementary groups without removing existing memberships.
- -G specifies the supplementary group.
- cybersecurity is the group name.
- intern1 is the username.

This command adds intern1 to the cybersecurity group.

Step 6: Verify User and Group Membership

Command:

id intern1

Description:

The id command displays the user's UID, primary GID, and supplementary group memberships.

Illustrative output:

uid=1001(intern1) gid=1001(intern1) groups=1001(intern1),1002(cybersecurity)

This confirms that intern1 belongs to the cybersecurity group in the example environment.

Step 7: Create Directory and Assign Ownership

Commands:

sudo mkdir /home/intern1/practice

sudo chown intern1:cybersecurity /home/intern1/practice

sudo chmod 770 /home/intern1/practice

Description:

These commands create a practice directory, assign ownership to user intern1 and group cybersecurity, and configure its access permissions.

The permission value 770 grants full access to the owner and group while denying access to others.

Step 8: Verify Directory Permissions

Command:

ls -ld /home/intern1/practice

Illustrative output:

drwxrwx--- 2 intern1 cybersecurity 4096 Oct 20 20:00 /home/intern1/practice

Explanation:

The output displays the directory's permission settings, owner, group, and other details.

Permission	Owner	Group	Others
Read	Yes	Yes	No
Write	Yes	Yes	No
Execute	Yes	Yes	No

Note: Access to this directory also depends on the permissions of its parent directories.

5. Linux Access Permissions

Linux provides three primary permission types: read, write, and execute.

Symbol	Permission	Numeric Value	Function
r	Read	4	Allows viewing file contents
w	Write	2	Allows modifying file contents
x	Execute	1	Allows executing files or traversing directories
-	None	0	No permission granted

Common Permission Settings

Permission	Meaning	Common Use
700	Full access for owner only	Private directories
750	Full owner access, group read/execute	Restricted shared directories
755	Owner full access, others read/execute	Publicly accessible directories
770	Full access for owner and group	Group collaboration
600	Owner read/write only	Sensitive files
644	Owner read/write, others read-only	General documents

6. Importance of Linux User Management

1. Access Control: Ensures users access only authorized files and directories.
2. Data Protection: Helps protect sensitive information from unauthorized modification.
3. User Accountability: Provides individual accounts for identifying user activities.
4. Group-Based Security: Simplifies permission management for multiple users.
5. System Administration: Allows administrators to create, modify, and manage accounts efficiently.
6. Least Privilege: Reduces security risks by limiting unnecessary permissions.

7. Security Best Practices

- Create separate accounts for individual users.
- Avoid using the root account for everyday activities.
- Assign only the permissions required for a user's responsibilities.
- Use strong passwords and secure authentication methods.
- Review user accounts and group memberships regularly.
- Remove or disable accounts that are no longer required.
- Use sudo only when administrative privileges are necessary.
- Restrict access to sensitive files and directories.

8. Results and Observations

The illustrated Linux user management workflow demonstrates the following operations.

Activity	Illustrated Result
Current user identification	Username displayed
User account listing	Existing accounts displayed
New user creation	intern1 account created
Group creation	cybersecurity group created
Group assignment	intern1 added to cybersecurity
Membership verification	UID, GID, and groups displayed
Directory ownership	Assigned to intern1: cybersecurity
Permission configuration	770 applied
Permission verification	drwxrwx--- displayed

The generated terminal screenshot provides an illustrative representation of the commands and expected outputs. These results should be verified in a real Linux environment before being described as executed test evidence.

9. Conclusion

The Linux User Management task demonstrated the essential concepts of user accounts, groups, file ownership, and access permissions in Linux.

Commands such as useradd, groupadd, usermod, id, chown, and chmod were explored to understand how administrators manage user access and protect system resources.

The task also highlighted the importance of group-based access control and the principle of least privilege.

Overall, this exercise strengthened the understanding of Linux account administration and its role in maintaining operating system security.

